



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The company experienced an ICMP flood denial-of-service attack that exploited an unconfigured firewall, causing all network services to stop responding and rendering internal network resources unavailable for approximately two hours. The cybersecurity team contained the incident by blocking the malicious ICMP traffic and shutting down noncritical services so that critical operations could be restored. Following the incident, the organization strengthened its defenses by implementing ICMP rate limiting, source IP verification to prevent spoofing, enhanced network monitoring, and an IDS/IPS to detect and filter suspicious traffic.
Identify	A malicious actor launched an ICMP flood denial-of-service (DoS) attack against the organization, overwhelming an unconfigured firewall and disrupting the entire internal network. The affected assets included the firewall, network infrastructure, and all network-dependent services and systems, rendering critical resources inaccessible until they could be secured and restored to normal operation.

Protect	The cybersecurity team strengthened the organization's defenses by implementing firewall rules to rate-limit incoming ICMP traffic and deploying an IDS/IPS to identify and filter suspicious packets. To further protect assets, the company should conduct regular firewall configuration reviews and security audits, maintain anti-spoofing controls, enforce secure configuration and change-management procedures, and provide ongoing network security and incident response training for staff.
Detect	The cybersecurity team enhanced detection capabilities by enabling source IP verification on the firewall to identify spoofed ICMP packets and deploying network-monitoring software to detect abnormal traffic patterns. To further improve detection, the organization should continuously monitor network activity with SIEM and network-monitoring tools, maintain IDS/IPS signatures for ICMP anomalies, regularly review firewall and system logs, and configure alerts for unusual traffic volumes and unauthorized access attempts.
Respond	For future security incidents, the cybersecurity team will follow a formal incident response playbook that includes isolating affected systems to prevent further disruption, filtering malicious traffic, and restoring critical services as quickly as possible. The team will preserve and analyze network logs and packet captures for forensic investigation, notify management and any required legal authorities, document lessons learned, and update security controls and response procedures based on the findings to strengthen the organization's resilience against future attacks.
Recover	To recover from an ICMP flood DDoS attack, the organization should follow a prioritized recovery plan that first blocks malicious ICMP traffic at the firewall

	and temporarily suspends nonessential services to reduce network load. Critical network services should then be restored and their integrity, availability, configurations, and backups verified before bringing noncritical systems back online once the attack has subsided. A post-incident review should be conducted afterward to improve business continuity and disaster recovery procedures for future events.
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Reflections/Notes: This incident underscores the importance of defense in depth: secure configurations, proactive monitoring, and a well-practiced incident response process significantly reduce the impact of DoS attacks.